

Solution to Problem set 5a: More practice of international trade

- 1) Suppose Italy has an hourly wage of 10 EUR per hour. Spain has an hourly wage of 15 EUR per hour. Furthermore, workers in Italy are twice as fast as workers in Spain. Each country can produce either Mangos or Trucks. In Italy, workers need 10 hours to produce one Mango. In the same time, they can instead produce 2 Trucks. There is perfect competition and the wage is the same in both the Mango and Truck industry in both countries.
- Will Italy export Trucks to Spain?
 - Italy decides to invest in a training program that will make its workers even faster, so that they will be 3 times (instead of just twice) as fast as in Spain. Will Italy export Trucks to Spain now?

(a) Let us construct a table, as in problem set 5.

For one item:	Italy	Spain
Mangos		
Hourly wage	10	15
Hours worked	10	20
Unit costs	100	300
Trucks		
Hourly wage	10	15
Hours worked	5	10
Unit costs	50	150

$$P_{Truck}^{IT} = \frac{50}{100} = \frac{1}{2}$$

$$P_{Truck}^{SP} = \frac{150}{300} = \frac{1}{2}$$

No country has a comparative advantage, and hence there is no trade (Italy does not export).

(b) After the change, we get:

For one item:	Italy	Spain
Mangos		
Hourly wage	10	15
Hours worked	10	30
Unit costs	100	450
Trucks		
Hourly wage	10	15
Hours worked	5	15
Unit costs	50	225

For simplicity, we don't need to shift the number in Italy, so again $P_{Truck}^{IT} = \frac{50}{100} = \frac{1}{2}$. For Spain, we now get $P_{Truck}^{SP} = \frac{225}{450} = \frac{1}{2}$. Now, again there is no comparative advantage and hence no trade. The reason is that while Italy is both faster and more efficient (and hence has an absolute advantage) there is no difference in the ratio of costs for the products. Hence, Italy does not have a comparative advantage. As a side note: a comparative advantage may arise if, e.g., the wage in the two industries would be different in each country. A

- 2) Malta and Canada are trying to decide whether to impose a tariff. If both of them impose a tariff, they each have a payoff of 1. If only one country imposes a tariff, then that country gets 10 and the other country gets 0. If no country imposes a tariff, they each get 6.
- a. Is this a prisoner's dilemma? Yes. There is a dominant strategy to impose a tariff, which yields a KH inefficient and Pareto inefficient outcome.

	Taarif	No taarif
Taarif	1,1	10,0
No Taarif	0, 10	6,6

- b. Does it matter whether they play this game 1 time or 50 times? *No. In a finitely repeated game, the result is the same as a one-shot game.*
- c. Does it matter whether they make the decision simultaneously, or whether one goes first and the other goes second (after observing the first)? *No. When there is a dominant strategy, the switch from a static (simultaneous) game to a dynamic game makes no difference.*
- d. Suppose the countries are considering to cooperate by using a grim-trigger strategy. What is the critical discount factor above which cooperation (i.e. no one imposes a tariff) can be sustained as an equilibrium?

$$\delta^{critical} = \frac{\pi_D - \pi_c}{\pi_D - \pi_p} = \frac{10 - 6}{10 - 1} = \frac{4}{9}$$

- a. Suppose that the discount factor is $\delta = \frac{1}{3}$. Is cooperation then possible as an equilibrium? *No. because it is less than the critical discount factor.*
- b. Suppose that the discount factor is $\delta = \frac{9}{10}$. Will there be cooperation then for sure, or is there another equilibrium as well? *The non-cooperative equilibrium is still a Nash equilibrium. Hence, cooperation is not the only equilibrium, even though this discount factor is above the critical discount factor.*
- c. Suppose that the parties can bargain at zero transaction costs. Would the equilibrium then change, and if so – what will it be? *The Coase theorem then suggests that they then reach an efficient solution and split the difference. Note: this means they choose the KH efficient outcome (as then surplus is then maximal). In this case, it is KH efficient if both do not impose a tariff ($6+6=12 > 10+0=10$). Hence, they cooperate and split the surplus (which is equal to $12 - 2 = 10$).*